



## Modelling Financial Markets

Seminar 7: Value-at-Risk and Expected Shortfall — Hands-on



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## Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CF: Cornish–Fisher (expansion)
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ DM: Diebold–Mariano (test)
- ▣ ES: Expected Shortfall
- ▣ EVT: Extreme Value Theory
- ▣ FHS: Filtered Historical Simulation
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GAS: Generalised Autoregressive Score (model)
- ▣ GPD: Generalised Pareto Distribution
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ LGD: Loss Given Default
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ QML: Quasi-Maximum Likelihood
- ▣ SE: Standard Error
- ▣ XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

# Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Questions

Wrap-up

References

## Seminar Objectives

### By the end of this seminar you can

- ▣ Derive VaR and ES for the Normal and Student- $t$  distributions and the sampling variance of their historical estimates
- ▣ Prove subadditivity of ES from the Rockafellar–Uryasev representation
- ▣ Estimate historical, filtered, extreme-value and CAViaR VaR/ES with honest intervals (HAC, block and residual bootstrap, declustering)
- ▣ Split portfolio risk into Euler contributions

- ▣ Background: Lecture 7; textbooks McNeil, Frey & Embrechts (2015) and Franke, Härdle & Hafner (2019)

### Structure

- ▣ **Part A:** derivations (A1–A8)
- ▣ **Part B:** seven computational exercises with inference (B1–B7)
- ▣ **Part C:** two open questions, seeds for the team project (C1, C3), and a critique of an AI answer (C2)

## How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
  - ▶ Every Part B exercise ends with interpretation questions
- Sign convention: loss  $L = -r$  in % of the position; VaR and ES are positive numbers
- Report every tail estimate with its uncertainty: how many observations lie beyond it?

## Solved and Proposed Problems

### Solved: worked examples in this deck

- Part A: A1 (Normal ES derivation, SE of historical VaR), A2 (Student- $t$  ES derivation), A3 (a discrete loss), A5 (component VaR; (d) ES contributions)
- Part B: B1 (historical VaR and ES on BET; bootstrap, asymptotic and HAC intervals), B2 (GPD fit on BET; extremal index)

### Proposed: you solve them

- Part A: A4 (failure of subadditivity; model: A3), A6 (validity of Cornish–Fisher; model: A1), A7 (ES as a minimum; model: A3, A4), A8 (variance of historical ES; model: A1)
- Part B: B3 (filtered vs historical on Bitcoin; model: B1), B4 (Cornish–Fisher vs empirical; model: A1, B1), B5 (component VaR of a portfolio; model: A5), B6 (bootstrap interval for ES; model: B1), B7 (CAViaR on BET; model: B3)
- Part C: C1, open question and project seed; C2, critique of an AI answer (model: B1); C3, dynamic (VaR, ES) models estimated by FZ0, project seed from the lecture case study

## Setup: Python Environment and Data

-  **Open in Colab:** `chapter7_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python  $\geq 3.10$  with `numpy`, `pandas`, `scipy`, `arch`, `matplotlib`
- Series: BET (2000–2026), S&P 500, Bitcoin (2014–2026), EUR/RON (BNR reference rate), gold; SPY, TLT, GLD; eight BVB blue chips; daily closes up to 18 September 2026
- Equity indices: weekdays only; holiday records (unchanged close, zero volume) removed, genuine unchanged closes kept as zero returns; crypto keeps all 7 days of the week
- Portfolios: prices joined on common trading days first, then simple returns

## Formulas Used Today

- Level  $\alpha$  = tail probability (VaR 1%, ES 2.5%); return  $X$ , loss  $L = -X$ :  $\text{VaR}_\alpha(X) = -\inf\{x : P(X \leq x) > \alpha\} = -q_\alpha(X)$ ,  
 $\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du$
- Discrete case [Acerbi & Tasche \(2002\)](#):  $\text{ES}_\alpha = \frac{1}{\alpha} (\mathbb{E}[L1\{L > \text{VaR}_\alpha\}] + \text{VaR}_\alpha(\alpha - P(L > \text{VaR}_\alpha)))$
- Normal distribution,  $z_\alpha = \Phi^{-1}(\alpha) < 0$ :  $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$ ,  $\text{ES}_\alpha = -\mu + \sigma \varphi(z_\alpha)/\alpha$
- Student- $t$ :  $\text{VaR}_\alpha = -(m + s t_\alpha)$ ,  $\text{ES}_\alpha = -m + s \frac{s\nu(t_\alpha)}{\alpha} \frac{\nu+t_\alpha^2}{\nu-1}$ ; variance  $\sigma^2$  when  $s = \sigma\sqrt{(\nu-2)/\nu}$
- GPD tail of the losses:  $\text{VaR}_\alpha = u + \frac{\beta}{\xi} \left[ \left( \frac{n}{N_u} \alpha \right)^{-\xi} - 1 \right]$ ,  $\text{ES}_\alpha = \frac{\text{VaR}_{\alpha+\beta-\xi u}}{1-\xi}$  ( $\alpha < N_u/n$ ; ES needs  $\xi < 1$ )
- Euler:  $C_i = w_i \partial \text{VaR} / \partial w_i = \mathbf{z}_{1-\alpha} w_i (\Sigma w)_i / \sigma_p$ ,  $\sum_i C_i = \text{VaR}$  [Tasche \(2008\)](#)

## A1: Normal VaR and ES [Solved]: Task

### □ Data and assumptions

- ▶ Daily returns of a position are Normal,  $X \sim N(\mu, \sigma^2)$ , with  $\mu = 0.04\%$  and  $\sigma = 1.2\%$ ; the loss is  $L = -X$ .
- ▶ We write  $z_\alpha = \Phi^{-1}(\alpha) < 0$ , and  $\Phi, \varphi$  for the standard Normal distribution and density functions.

### □ Tasks

- ▶ (a) Derive  $ES_\alpha = -\mu + \sigma\varphi(z_\alpha)/\alpha$  starting from the definition  $ES_\alpha = -\mathbb{E}[X \mid X \leq q_\alpha(X)]$ .
- ▶ (b) Compute VaR and ES at  $\alpha = 1\%$  and at  $\alpha = 2.5\%$ , in % of the position.
- ▶ (c) Compute the asymptotic standard error of the *historical* VaR 1%,  $SE = \sqrt{\alpha(1-\alpha)/n}/f(\text{VaR}_\alpha)$ , for  $n = 500$  and for  $n = 6\,670$  days, and the width of the 95% interval.

### □ Report

- ▶ The four numbers of (b), the two standard errors of (c), and a comparison with the bootstrap interval widths of B1: why are the B1 intervals wider?

## A1: Normal VaR and ES [Solved]: Solution (1/2)

- **(a) Derivation.** Write  $X = \mu + \sigma Z$ ; the event  $X \leq q_\alpha$  is  $Z \leq z_\alpha$ , which has probability  $\alpha$ .
  - ▶ Since  $\varphi'(u) = -u\varphi(u)$ , we get  $\int_{-\infty}^z u\varphi(u) du = [-\varphi(u)]_{-\infty}^z = -\varphi(z)$ .
  - ▶ Hence  $\mathbb{E}[X \mid X \leq q_\alpha] = \mu + \frac{\sigma}{\alpha} \int_{-\infty}^{z_\alpha} u\varphi(u) du = \mu - \sigma \frac{\varphi(z_\alpha)}{\alpha}$ , and changing the sign gives the formula.
- **(b) Numbers at 1%.** The quantile is  $z_{1\%} = -2.326$  and the density there is  $\varphi(z_{1\%}) = 0.0267$ , so  $\varphi/\alpha = 2.665$ .
  - ▶  $\text{VaR}_{1\%} = -(0.04 + 1.2 \cdot (-2.326)) = 2.75\%$  and  $\text{ES}_{1\%} = -0.04 + 1.2 \cdot 2.665 = 3.16\%$ .
- **(b) Numbers at 2.5%.** Here  $z_{2.5\%} = -1.960$  and  $\varphi/\alpha = 2.338$ .
  - ▶  $\text{VaR}_{2.5\%} = 2.31\%$  and  $\text{ES}_{2.5\%} = -0.04 + 1.2 \cdot 2.338 = 2.77\%$ .
  - ▶ Interpretation: ES 2.5% (2.77%) is almost equal to VaR 1% (2.75%); this near-equality under the Normal distribution is why FRTB chose ES 2.5%.

## A1: Normal VaR and ES [Solved]: Solution (2/2)

- **(c) Density at the quantile.** The loss density at VaR 1% is  $f(\text{VaR}_{1\%}) = \varphi(z_{1\%})/\sigma = 0.0267/1.2 = 0.0222$ .
  - ▶  $n = 500$ :  $\text{SE} = \sqrt{0.01 \cdot 0.99/500}/0.0222 = 0.20$  percentage points, so the 95% interval is 0.79 points wide.
  - ▶  $n = 6\,670$ :  $\text{SE} = 0.055$  and the 95% width is 0.22 points.
- **Comparison with B1.** The bootstrap intervals for BET VaR 1% are 1.01 points wide (500 days) and 0.61 points wide (full sample).
  - ▶ Interpretation: BET losses have a heavier tail than the Normal distribution, so the density at VaR is lower and the quantile is estimated less precisely.
  - ▶ The standard error falls like  $1/\sqrt{n}$ : thirteen times more data make the interval about  $\sqrt{13.3} \approx 3.7$  times narrower.

## A2: Student- $t$ ES [Solved]: Task

### □ Data and assumptions

- ▶ Returns are  $X = m + sT$  with  $T$  Student- $t$  with  $\nu$  degrees of freedom and density  $g_\nu(x) = c_\nu(1 + x^2/\nu)^{-(\nu+1)/2}$ .
- ▶ The quantile is  $t_\alpha = t_\nu^{-1}(\alpha) < 0$ ; for the numbers use  $\nu = 4$ ,  $m = 0.04\%$  and the same standard deviation 1.2% as in A1.

### □ Tasks

- ▶ (a) Show that  $\int_{-\infty}^t x g_\nu(x) dx = -g_\nu(t) \frac{\nu + t^2}{\nu - 1}$  for  $\nu > 1$ , and deduce  $\text{ES}_\alpha = -m + s \frac{g_\nu(t_\alpha)}{\alpha} \frac{\nu + t_\alpha^2}{\nu - 1}$ .
- ▶ (b) Find the scale  $s$  that gives a standard deviation of 1.2%, using  $\text{Var}(T) = \nu/(\nu - 2)$ .
- ▶ (c) Compute VaR and ES at  $\alpha = 1\%$  and  $\alpha = 2.5\%$ .

### □ Report

- ▶ The four numbers of (c) next to those of A1; interpretation question: at which level do the two models differ most, and why?

## A2: Student- $t$ ES [Solved]: Solution

- (a) **Antiderivative.** Differentiate  $(1 + x^2/\nu)^{-(\nu-1)/2}$ : the result is  $-\frac{\nu-1}{\nu} x (1 + x^2/\nu)^{-(\nu+1)/2}$ .
  - ▶ Therefore  $xg_\nu(x) = -\frac{\nu c_\nu}{\nu-1} \frac{d}{dx} (1 + x^2/\nu)^{-(\nu-1)/2}$ , and integrating from  $-\infty$  to  $t$  gives  $-\frac{\nu}{\nu-1} g_\nu(t)(1 + t^2/\nu)$ .
  - ▶ This equals  $-g_\nu(t)(\nu + t^2)/(\nu-1)$ ; the limit at  $-\infty$  vanishes only if  $\nu > 1$ . Divide by  $\alpha$  and rescale by  $m + sT$  as in A1.
- (b) **Scale.**  $s^2\nu/(\nu-2) = 1.2^2$ , so  $s = 1.2\sqrt{2/4} = 0.849$ .
- (c) **At 1%.**  $t_{1\%} = -3.747$  and  $g_4(t_{1\%}) = 0.0087$ , so the ES factor is  $\frac{0.0087}{0.01} \cdot \frac{4 + 14.04}{3} = 5.221$ .
  - ▶  $\text{VaR}_{1\%} = -(0.04 + 0.849 \cdot (-3.747)) = 3.14\%$  and  $\text{ES}_{1\%} = -0.04 + 0.849 \cdot 5.221 = 4.39\%$ .
- (c) **At 2.5%.**  $t_{2.5\%} = -2.776$  gives  $\text{VaR}_{2.5\%} = 2.32\%$  and  $\text{ES}_{2.5\%} = 3.35\%$ .
  - ▶ Interpretation: VaR 2.5% is almost the same in both models (2.32 vs 2.31), while VaR 1% and both ES values are much higher under  $t_4$ : heavy tails show only deep in the tail, and ES sees them first.

## A3: VaR and ES of a Discrete Loss [Solved]: Task

### □ Data and assumptions

- ▶ A bond loses 60 (loss given default, LGD) with probability  $p = 3\%$  and 0 otherwise, so the return is  $X = -60$  or  $X = 0$ .
- ▶ Use the general definitions  $\text{VaR}_\alpha = -\inf\{x : P(X \leq x) > \alpha\}$  and the discrete ES formula of the lecture [Acerbi & Tasche \(2002\)](#).

### □ Tasks

- ▶ (a) Compute VaR 5% and VaR 2.5% directly from the definition.
- ▶ (b) Compute ES 5% and ES 2.5% with  $\text{ES}_\alpha = \frac{1}{\alpha} (\mathbb{E}[L1\{L > \text{VaR}_\alpha\}] + \text{VaR}_\alpha(\alpha - P(L > \text{VaR}_\alpha)))$ .
- ▶ (c) Explain why  $\text{ES}_{5\%}$  is not equal to  $\mathbb{E}[L \mid L \geq \text{VaR}_{5\%}]$ .

### □ Report

- ▶ The four numbers and a one-sentence answer to (c).

### A3: VaR and ES of a Discrete Loss [Solved]: Solution

- (a) **VaR 5%.**  $P(X \leq -60) = 3\%$  is not above 5%, while  $P(X \leq 0) = 100\% > 5\%$ ; so  $q_{5\%} = 0$  and  $\text{VaR}_{5\%} = 0$ .
  - ▶ VaR 2.5%:  $P(X \leq -60) = 3\% > 2.5\%$ , so  $q_{2.5\%} = -60$  and  $\text{VaR}_{2.5\%} = 60$ .
- (b) **ES 5%.**  $\mathbb{E}[L1\{L > 0\}] = 60 \cdot 0.03 = 1.8$  and  $P(L > 0) = 3\%$ , so  $\text{ES}_{5\%} = \frac{1.8 + 0 \cdot (0.05 - 0.03)}{0.05} = 36$ .
  - ▶ ES 2.5%: nothing lies above 60, so  $\text{ES}_{2.5\%} = \frac{0 + 60 \cdot (0.025 - 0)}{0.025} = 60$ .
- (c) **Why the correction term.**  $\mathbb{E}[L \mid L \geq 0] = 1.8$  averages over **all** days, not over the worst 5%.
  - ▶ Interpretation: with atoms, the worst 5% of outcomes consist of the 3% default mass plus 2% of the zero-loss mass; the formula weights them exactly, and ES then equals 36.

## A4: When VaR Punishes Diversification [Proposed]: Task

Solution discussed in class

### □ Data and assumptions

- ▶ Two independent bonds A and B; each loses 100 with probability  $p = 0.7\%$  and 0 otherwise [Artzner et al. \(1999\)](#). The portfolio is  $A + B$ .

### □ Tasks

- ▶ (a) Write the distribution of the portfolio loss (values 0, 100, 200 and their probabilities).
- ▶ (b) Compute VaR 1% and ES 1% of one bond and of the portfolio, with the discrete formula of A3.
- ▶ (c) Show that VaR 1% is not subadditive here while ES 1% is.
- ▶ (d) Repeat (b)–(c) at  $\alpha = 1.5\%$  and find the range of  $\alpha$  for which VaR fails.

### □ Report

- ▶ A small table of VaR and ES for A, B and  $A + B$  at both levels, and one sentence on what a VaR-based limit would reward.
- ▶ Model: A3.

## A5: Component VaR and ES of Two Assets [Solved]: Task

### □ Data and assumptions

- ▶ Equity 600 000 EUR with  $\sigma_1 = 1.2\%$  daily; bonds 400 000 EUR with  $\sigma_2 = 0.9\%$ ; correlation  $\rho = -0.2$ ; Normal distribution, zero mean,  $\alpha = 1\%$ .

### □ Tasks

- ▶ (a) Compute the portfolio standard deviation in EUR and the portfolio VaR 1%.
- ▶ (b) Compute the marginal VaR  $z_{1-\alpha}(\Sigma w)_i/\sigma_p$  and the component VaR  $C_i$  of each asset, and check  $C_1 + C_2 = \text{VaR}$  Tasche (2008).
- ▶ (c) Compute the diversification benefit (sum of stand-alone VaRs minus portfolio VaR).
- ▶ (d) Using  $\text{ES}_\alpha(w) = \min_v \{v + \mathbb{E}[(L_p - v)_+]/\alpha\}$  Rockafellar & Uryasev (2000), show that  $w_i \partial \text{ES}_\alpha / \partial w_i = \mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_\alpha]$  and evaluate these ES contributions here.

### □ Report

- ▶ A table of  $C_i$  and  $C_i^{ES}$  with shares; interpretation: do the bonds carry risk in proportion to their money weight?

## A5: Component VaR and ES of Two Assets [Solved]: Solution (1/2)

- (a) **Portfolio risk.** Stand-alone standard deviations:  $600\,000 \cdot 1.2\% = 7\,200$  EUR and  $400\,000 \cdot 0.9\% = 3\,600$  EUR.
  - ▶  $\sigma_p^2 = 7\,200^2 + 3\,600^2 + 2(-0.2)(7\,200)(3\,600) = 54\,432\,000$ , so  $\sigma_p = 7\,378$  EUR and  $\text{VaR}_{1\%} = 2.326 \cdot 7\,378 = 17\,163$  EUR.
- (b) **Marginal and component VaR.**  $(\Sigma w) = (77.76, 19.44)$  (in EUR, with  $\Sigma$  in decimal units), which gives the marginal VaR per EUR of exposure  $(0.0245, 0.0061)$ .
  - ▶  $C_1 = 14\,711$  EUR (85.7%) and  $C_2 = 2\,452$  EUR (14.3%); their sum is 17 163 EUR, exactly the portfolio VaR.
- (c) **Diversification.** Stand-alone VaRs  $16\,750 + 8\,375 = 25\,125$  EUR against a portfolio VaR of 17 163 EUR.
  - ▶ Interpretation: the bonds hold 40% of the money but only 14.3% of the risk, because the negative correlation makes them a partial hedge.

## A5: Component VaR and ES of Two Assets [Solved]: Solution (2/2)

- **(d) Envelope argument.**  $v^* = \text{VaR}_\alpha$  minimises  $G(w, v) = v + \mathbb{E}[(L_p - v)_+]/\alpha$ , so  $\partial \text{ES}_\alpha / \partial w_i = \partial G / \partial w_i$  at  $v^*$ .
  - ▶ Differentiating under the expectation:  $\partial G / \partial w_i = \mathbb{E}[-R_i \mathbf{1}\{L_p > v^*\}]/\alpha$ .
  - ▶ For a continuous  $L_p$ ,  $P(L_p > \text{VaR}_\alpha) = \alpha$ , so this is  $\mathbb{E}[-R_i \mid L_p \geq \text{VaR}_\alpha]$ ; multiplying by  $w_i$  and summing gives  $\mathbb{E}[L_p \mid L_p \geq \text{VaR}_\alpha] = \text{ES}_\alpha$ .
- **(d) Numbers.** Under the Normal distribution  $\text{ES}_\alpha = \sigma_p \varphi(z_\alpha) / \alpha$ , so every contribution scales by  $\frac{\varphi(z_\alpha) / \alpha}{z_{1-\alpha}} = 1.146$ .
  - ▶  $C_1^{\text{ES}} = 16\,854$  EUR and  $C_2^{\text{ES}} = 2\,809$  EUR; together 19 663 EUR, the portfolio ES 1%.
  - ▶ Interpretation: under the Normal distribution the ES shares equal the VaR shares, so ES adds no new allocation information; with heavy tails it does (see B5).

## A6: When Is Cornish–Fisher a Valid Quantile? [Proposed]: Task

Solution discussed in class

### □ Data and assumptions

- ▶ Returns with  $\mu = 0$ ,  $\sigma = 1.2\%$ , skewness  $S = -0.5$  and excess kurtosis  $K$  [Cornish & Fisher \(1938\)](#).
- ▶ The Cornish–Fisher map is  $\tilde{z}(z) = z + \frac{(z^2 - 1)S}{6} + \frac{(z^3 - 3z)K}{24} - \frac{(2z^3 - 5z)S^2}{36}$  and  $\text{VaR}_\alpha = -\sigma\tilde{z}(z_\alpha)$ .

### □ Tasks

- ▶ (a) With  $K = 3$ , compute the three correction terms at  $z_{1\%}$ , then  $\tilde{z}_{1\%}$  and the Cornish–Fisher VaR 1%; compare with the Normal VaR.
- ▶ (b) Derive  $d\tilde{z}/dz$  and find where it is smallest on  $[z_{0.1\%}, 0]$ .
- ▶ (c) Find the largest  $K$  for which  $\tilde{z}$  is increasing on this interval, and decide whether  $K = 10$  (close to the S&P 500) is admissible.

### □ Report

- ▶ The VaR of (a), the bound on  $K$ , and whether a VaR computed with  $K = 10$  is a valid estimate. Model: A1.

## A7: ES as a Minimum, and Subadditivity [Proposed]: Task

Solution discussed in class

### □ Data and assumptions

- ▶  $L$  is any integrable loss; define  $F_L(v) = v + \mathbb{E}[(L - v)_+]/\alpha$  Rockafellar & Uryasev (2000). For the numbers, use the A4 portfolio (two bonds,  $p = 0.7\%$ ) at  $\alpha = 1\%$ .

### □ Tasks

- ▶ (a) Prove that  $F_L$  is convex and that its minimum over  $v$  equals  $\text{ES}_\alpha(L)$ , attained at  $v = \text{VaR}_\alpha(L)$ .
- ▶ (b) Deduce  $\text{ES}_\alpha(L_1 + L_2) \leq \text{ES}_\alpha(L_1) + \text{ES}_\alpha(L_2)$  from  $(a + b)_+ \leq a_+ + b_+$ .
- ▶ (c) For the A4 portfolio, compute the slope of  $F$  on  $(0, 100)$  and on  $(100, 200)$ , the minimiser and the minimum.

### □ Report

- ▶ The two proofs in a few lines and the three numbers of (c), compared with  $\text{ES}_{1\%}(A + B)$  from A4. Model: A3, A4.

## A8: The Sampling Variance of Historical ES [Proposed]: Task

Solution discussed in class

### □ Data and assumptions

- ▶  $n$  i.i.d. losses with a continuous distribution, a positive density at  $\text{VaR}_\alpha$  and  $\mathbb{E}[L^2] < \infty$ ; historical ES is the mean of the losses beyond the empirical VaR. Its influence function is  $\text{IF}(x) = \text{VaR}_\alpha + (x - \text{VaR}_\alpha)_+ / \alpha - \text{ES}_\alpha$  [Chen \(2008\)](#).

### □ Tasks

- ▶ (a) Show that the variance of the limit of  $\sqrt{n}(\widehat{\text{ES}}_\alpha - \text{ES}_\alpha)$  is  $\text{avar}(\widehat{\text{ES}}_\alpha) = \text{Var}(\text{IF}(L)) = \text{Var}((L - \text{VaR}_\alpha)_+) / \alpha^2$ , and explain why estimating VaR adds no term.
- ▶ (b) For  $L \sim N(0, 1)$  and  $\alpha = 2.5\%$ , compute  $\mathbb{E}[(L - v)_+]$  and  $\mathbb{E}[(L - v)_+^2]$  in closed form at  $v = \text{VaR}_\alpha$ .
- ▶ (c) Compute the standard error of  $\widehat{\text{ES}}_{2.5\%}$  for  $n = 500$ , for  $\sigma = 1$  and for  $\sigma = 1.2\%$ , and compare with the S&P 500 over its last 500 days.

### □ Report

- ▶ The two moments, the asymptotic variance and the two standard errors; interpretation: why is the S&P 500 value much larger? Model: A1.

## B1: Historical VaR and ES on BET [Solved]: Task

### □ Data and assumptions

- ▶ BET index (Bucharest Stock Exchange), data/market/BET.csv, daily closes 6 January 2000 – 18 September 2026; daily log returns in %,  $r_t = 100 (\ln P_t - \ln P_{t-1})$ , loss  $L = -r$ ; weekdays only (the file has no holiday records).
- ▶ Two samples: the full period ( $n = 6\,670$ ) and the last 500 days (from 2024-09-12).

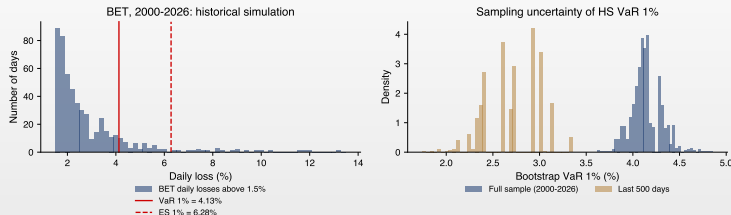
### □ Tasks

- ▶ (a) Compute historical VaR and ES at  $\alpha = 1\%$  and  $\alpha = 2.5\%$  on both samples (VaR = empirical  $(1 - \alpha)$  quantile of  $L$ , ES = mean of the losses at or above it).
- ▶ (b) Build a 95% percentile bootstrap interval for VaR 1% with 2000 i.i.d. resamples [Efron \(1979\)](#).
- ▶ (c) Build three more 95% intervals for VaR 1%: asymptotic i.i.d. ( $\widehat{\text{VaR}} \pm 1.96 \sqrt{\alpha(1 - \alpha)/n/\hat{f}}$ , Gaussian kernel density, Silverman bandwidth), asymptotic HAC (Bartlett kernel, lag  $\lfloor 4(n/100)^{2/9} \rfloor$ ), and a moving-block bootstrap with 20-day blocks [Künsch \(1989\)](#).

### □ Report

- ▶ A table with the eight VaR/ES numbers and the four intervals. Interpretation: how many losses lie beyond VaR 1% in each sample, which estimate would you report to a risk committee, and which interval is widest and why?

## B1: Historical VaR and ES on BET [Solved]: Solution (1/3)



|               | $n$   | $\text{VaR}_{1\%}$ | $\text{ES}_{1\%}$ | $\text{VaR}_{2.5\%}$ | $\text{ES}_{2.5\%}$ |
|---------------|-------|--------------------|-------------------|----------------------|---------------------|
| 2000–2026     | 6 670 | 4.13               | 6.28              | 2.80                 | 4.57                |
| Last 500 days | 500   | 2.71               | 3.22              | 2.05                 | 2.69                |

- (a) **Procedure.** Sort the losses, take the 99% (97.5%) empirical quantile as VaR and average the losses at or above it for ES; the chart shows the tail and the bootstrap distributions of VaR 1%.

## B1: Historical VaR and ES on BET [Solved]: Solution (2/3)

- (a) **Tail counts.** Beyond VaR 1% lie 67 losses in the full sample but only 5 in the last 500 days.
  - ▶ So the 500-day ES 1% is the average of only five numbers and should be reported with a warning.
- (b) **Bootstrap intervals.** Full sample [3.85, 4.46]; last 500 days [2.11, 3.12].
  - ▶ The two intervals do not overlap: the recent market is calmer than the 26-year average.
  - ▶ Interpretation: report the recent value for today's risk and the long-run value as a stress reference.

## B1: Historical VaR and ES on BET [Solved]: Solution (3/3)

|               | i.i.d. bootstrap | asympt., i.i.d. | asympt., HAC | block bootstrap |
|---------------|------------------|-----------------|--------------|-----------------|
| 2000–2026     | [3.85, 4.46]     | [3.81, 4.44]    | [3.69, 4.57] | [3.61, 4.80]    |
| Last 500 days | [2.11, 3.12]     | [2.18, 3.25]    | [2.19, 3.23] | [2.02, 3.03]    |

- **(c) Asymptotic step.**  $\hat{f}(\widehat{\text{VaR}}) = 0.0076$  (full) and  $0.0163$  (500 days), so  $\text{SE} = \sqrt{0.01 \cdot 0.99/n/\hat{f}} = 0.16$  and  $0.27$ ; the HAC standard errors are  $0.23$  and  $0.27$ .
- **Reading the table.** Full-sample widths:  $0.61, 0.63, 0.88, 1.19$ ; exceedances cluster in 2008 and 2020, so HAC and blocks widen the interval by a half to a factor of two.
- Last 500 days: all four widths are about 1; the period is calm, with little clustering, so the i.i.d. answer is adequate there.

## B2: GPD Fit and Extreme-Value VaR on BET [Solved]: Task

### □ Data and assumptions

- ▶ BET index (Bucharest Stock Exchange), data/market/BET.csv, daily closes 6 January 2000 – 18 September 2026; daily log returns in %,  $r_t = 100 (\ln P_t - \ln P_{t-1})$ , loss  $L = -r$ ; weekdays only (the file has no holiday records).
- ▶ Peaks over threshold: the excesses  $L - u$  above a high threshold  $u$  follow a generalised Pareto distribution (GPD) with shape  $\xi$  and scale  $\beta$  [McNeil, Frey & Embrechts \(2015\)](#).

### □ Tasks

- ▶ (a) Set  $u$  so that the largest 5% of losses lie above it and fit the GPD to the excesses by maximum likelihood.
- ▶ (b) Compute the i.i.d. standard errors  $SE(\hat{\xi}) = (1 + \hat{\xi})/\sqrt{N_u}$  and  $SE(\hat{\beta}) = \hat{\beta}\sqrt{2(1 + \hat{\xi})/N_u}$ .
- ▶ (c) Compute the GPD VaR and ES at  $\alpha = 1\%$ ,  $0.5\%$  and  $0.1\%$  and compare with historical simulation.
- ▶ (d) Refit for thresholds leaving from 10% down to 1.5% of the losses above them and plot  $\hat{\xi}$  with 95% bands.
- ▶ (e) Estimate the extremal index  $\theta$  of the exceedances of  $u$  with the intervals estimator [Ferro & Segers \(2003\)](#), decluster, and refit the GPD on the cluster maxima.

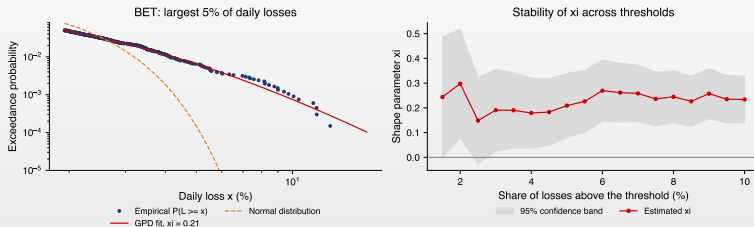
### □ Report

- ▶ The fit, the VaR/ES table, the stability chart and the declustered fit. Interpretation: is the BET tail heavier than the Normal distribution, how sure are we, and is the i.i.d. SE of  $\hat{\xi}$  credible?

## B2: GPD Fit and Extreme-Value VaR on BET [Solved]: Solution (1/3)

- (a) **Fit.**  $u = 1.93\%$  leaves  $N_u = 334$  of  $n = 6\,670$  losses above it; maximum likelihood gives  $\hat{\xi} = 0.209$  and  $\hat{\beta} = 1.20$ .
- (b) **Standard errors.**  $SE(\hat{\xi}) = (1 + 0.209)/\sqrt{334} = 0.066$  and  $SE(\hat{\beta}) = 0.10$ .
  - ▶  $\hat{\xi}/SE = 3.2$ : under independence the tail is significantly heavier than exponential (Normal-type); (e) checks the independence assumption.
- (c) **VaR 1%, step by step.**  $N_u/n = 0.050$ , so  $(0.01/0.050)^{-0.209} = 1.401$ .
  - ▶  $VaR_{1\%} = 1.93 + \frac{1.20}{0.209}(1.401 - 1) = 4.22\%$  (historical 4.13%);  $ES_{1\%} = 6.34\%$  (historical 6.28%).
  - ▶ Deeper: VaR 0.5% 5.47 vs 5.38; VaR 0.1% 9.18 vs 9.50, with only 7 observations beyond; ES 0.1% 12.61 vs 11.25.
  - ▶ Interpretation: the two methods agree down to 0.5%; at 0.1% historical ES averages seven numbers, and only the GPD gives a stable value.

## B2: GPD Fit and Extreme-Value VaR on BET [Solved]: Solution (2/3)



- **(d) Stability.** Across thresholds  $\hat{\xi}$  ranges from 0.15 to 0.30, but VaR 0.1% only from 9.04% to 9.51%.
- Reading the chart: left, the GPD follows the empirical tail and the Normal tail falls far below it; right, the band widens as fewer losses remain. The quantile is more stable than the shape parameter.

## B2: GPD Fit and Extreme-Value VaR on BET [Solved]: Solution (3/3)

- (e) **Extremal index.** From the gaps  $T_i$  between the 334 exceedances:  $\hat{\theta} = \frac{2(\sum_i (T_i - 1))^2}{(N - 1) \sum_i (T_i - 1)(T_i - 2)} = 0.32$ .
  - ▶ Declustering [Ferro & Segers \(2003\)](#): target  $C = \lfloor \hat{\theta}N \rfloor + 1 = 108$  clusters, cut at the  $C - 1$  largest gaps; the  $(C - 1)$ -th and  $C$ -th largest gaps are equal, so  $C$  is reduced until they differ, leaving 100 clusters of 3.3 exceedances on average.
  - ▶ At the VaR 1% threshold  $\hat{\theta} = 0.40$ .
- **Declustered fit.** GPD on the 100 cluster maxima:  $\hat{\xi} = 0.269$  (SE 0.127) against 0.209 (SE 0.066) on all exceedances.
  - ▶  $\hat{\xi}/\text{SE}$  falls from 3.2 to 2.1: only about 100 of the 334 excesses carry independent information.
  - ▶ Interpretation: the tail is still heavy, but the i.i.d. SE of 0.066 is not credible; report the declustered fit, or filter with GARCH first (conditional EVT).

## B3: Filtered vs Historical Simulation on Bitcoin [Proposed]: Task

Solution discussed in class

### □ Data and assumptions

- ▶ Bitcoin in USD, data/market/BTC-USD.CC.csv, daily closes 17 September 2014 – 18 September 2026, 7 days a week; daily log returns in %.
- ▶ Model:  $r_t = \mu_t + \sigma_t z_t$  with AR(1) mean and GARCH(1,1) variance, estimated by quasi-maximum likelihood Barone-Adesi et al. (1999).

### □ Tasks

- ▶ (a) Estimate the model on the full sample and forecast  $\hat{\mu}_{t+1}$  and  $\hat{\sigma}_{t+1}$  for 19 September 2026.
- ▶ (b) Compute the filtered VaR 1% and ES 1% from the empirical quantile of the standardised residuals, and compare with historical simulation on 500 days and on the full sample.
- ▶ (c) Using one-day-ahead forecasts (GARCH re-estimated each January), count the days of the last two years with a loss above each VaR.
- ▶ (d) Build a 90% residual-bootstrap interval for today's filtered VaR 1% with 999 paths, re-estimating the GARCH model on each path Christoffersen & Gonçalves (2005).

### □ Report

- ▶ The three VaR/ES pairs, the exceedance counts and the interval. Interpretation: why can the three VaRs differ by a factor of two, which method misses more often, and does the interval of (d) overlap the 500-day historical VaR? Model: B1.

## B4: Cornish–Fisher vs Empirical Quantiles [Proposed]: Task

Solution discussed in class

### □ Data and assumptions

- ▶ Daily log returns in % (data/market): S&P 500 and gold 2000–2026, BET 2000–2026, Bitcoin 2014–2026, EUR/RON (BNR reference rate) 2005–2026; full samples.

### □ Tasks

- ▶ (a) Compute the sample skewness  $S$  and excess kurtosis  $K$  of the returns of each series.
- ▶ (b) Compute VaR 1% and VaR 2.5% three ways: Normal, Cornish–Fisher (formula of A6) and historical.
- ▶ (c) Build a 95% percentile bootstrap interval (1 000 i.i.d. resamples) for the historical VaR 1% and check whether the Cornish–Fisher value lies inside.

### □ Report

- ▶ One table with all numbers. Interpretation: is Cornish–Fisher inside the interval for any series, why does it work better at 2.5% than at 1%, and would moments from the last year help? Model: A1, A6, B1.

## B5: Component VaR of a Four-Asset Portfolio [Proposed]: Task

Solution discussed in class

### □ Data and assumptions

- ▶ SPY, TLT, GLD (adjusted closes) and Bitcoin, data/market, 18 September 2014 – 18 September 2026; prices joined on the 3017 common days first, then daily simple returns in %.
- ▶ Equal weights of 25%, rebalanced daily;  $\alpha = 1\%$  for VaR and  $\alpha = 2.5\%$  for ES.

### □ Tasks

- ▶ (a) Compute the Normal VaR 1%  $z_{1-\alpha} \sqrt{w^\top \Sigma w}$ , the marginal VaR and the component VaR of each asset [Tasche \(2008\)](#).
- ▶ (b) Compute the historical ES 2.5% of the portfolio and the contributions  $\mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_{2.5\%}]$  as averages over the tail days.
- ▶ (c) Compute the undiversified VaR  $\sum_i z_{1-\alpha} w_i \sigma_i$ .

### □ Report

- ▶ A table of contributions and shares. Interpretation: is an equally weighted portfolio equally risky in each asset, how does it compare with the 40/30/20/10 portfolio of the lecture, and which weights would equalise the contributions? Model: A5.

## B6: How Precise Is an ES Estimate? [Proposed]: Task

Solution discussed in class

### □ Data and assumptions

- ▶ S&P 500 (data/market/GSPC.INDX.csv) and BET, daily log returns in %, 2000–2026; two samples each: full and last 500 days.

### □ Tasks

- ▶ (a) Compute historical ES 2.5% (lecture convention), the tail count  $n\alpha$  and the number  $k$  of losses the tail mean actually averages.
- ▶ (b) Build a 95% percentile interval with the i.i.d. bootstrap [Efron \(1979\)](#), 2 000 resamples.
- ▶ (c) Repeat (b) with the moving-block bootstrap, blocks of 20 days [Künsch \(1989\)](#).

### □ Report

- ▶ A table with  $n\alpha$ , ES and both intervals. Interpretation: why is the block interval wider on the full sample, and does a 500-day unconditional ES estimate pin ES down to within 0.3 percentage points [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)? Model: B1.

## B7: CAViaR on BET [Proposed]: Task

Solution discussed in class

### □ Data and assumptions

- ▶ BET index (Bucharest Stock Exchange), data/market/BET.csv, daily closes 6 January 2000 – 18 September 2026; daily log returns in %,  $r_t = 100 (\ln P_t - \ln P_{t-1})$ , loss  $L = -r$ ; weekdays only (the file has no holiday records).
- ▶ CAViaR with symmetric absolute value:  $\text{VaR}_t = \beta_0 + \beta_1 \text{VaR}_{t-1} + \beta_2 |r_{t-1}|$ ,  $\alpha = 1\%$ ,  $\text{VaR}_1 = -\hat{q}_\alpha$  of the first 300 returns (their upper loss quantile), these days kept in the estimation sample [Engle & Manganeli \(2004\)](#).

### □ Tasks

- ▶ (a) Estimate  $\beta$  on 2000–2015 by minimising the pinball loss  $\sum_t (\alpha - 1\{r_t < -\text{VaR}_t\})(r_t + \text{VaR}_t)$ :  $10^4$  random starts in  $(0, 1)^3$ , the best 10 refined by simplex and quasi-Newton steps.
- ▶ (b) With  $\beta$  fixed, forecast VaR for 2016–2026 and compute the exceedance rate and the mean pinball loss; compare with FHS (GARCH re-estimated each January) and 500-day historical simulation.
- ▶ (c) For 20 February – 30 April 2020 count the exceedances and report the peak VaR of each method.

### □ Report

- ▶ The estimates, a comparison table and the COVID-19 numbers. Interpretation: what does  $\hat{\beta}_1$  say about the memory of BET risk, how does CAViaR react to the crash compared with FHS, and are the exceedance counts different? (formal tests in Chapter 8). Model: B3; lecture, CAViaR on the S&P 500.

## C1: ES Capital, BVB vs S&P 500 [Open]: Task

### □ Data and assumptions

- ▶ BVB basket: Banca Transilvania, OMV Petrom, BRD, Transgaz, Romgaz, Nuclearelectrica, Electrica, Transelectrica (data/market, adjusted closes), equal weights, rebalanced daily; S&P 500 portfolio: SPY.
- ▶ Last ten years (from 2016-09-20), daily simple returns  $R_t$  in % in local currency, 1 million invested; a simplified illustration inspired by FRTB [Basel Committee \(2019\)](#), not the full capital calculation of [Basel Committee, MAR33](#).

### □ Tasks

- ▶ (a) Compute the one-day historical ES 2.5% of each portfolio (loss  $-R_t$ ).
- ▶ (b) Compute the 10-day ES 2.5% three ways:  $\sqrt{10}$  scaling; overlapping compounded losses  $100 [1 - \prod_{i=1}^{10} (1 + R_{t+i}/100)]$ ; filtered Monte Carlo from today's AR(1)-GARCH(1,1) state on the log returns, each path converted to a simple loss.
- ▶ (c) Stress window: the 250 consecutive days with the largest one-day historical ES 2.5%; compute the stressed ES inside it at 1 day and, by  $\sqrt{10}$  scaling, at 10 days.
- ▶ (d) Capital per million =  $m_c \times$  stressed ES,  $m_c = 1.5$ , for a liquidity horizon of 10 days (large cap) and of 20 days (small cap, a further factor  $\sqrt{20/10}$ ).
- ▶ (e) 95% moving-block bootstrap intervals (20-day blocks, 2000 resamples) for (a) and for the one-day stressed ES of (c), with the stress window held fixed.

### □ Report

- ▶ A table with all numbers and the intervals of (e), and a one-page argument: which portfolio needs more capital, and how do thin trading and currency (RON vs USD) change the answer?

## C2: Critique an AI Answer [Proposed]

Solution discussed in class

### The prompt sent to an artificial intelligence (AI) assistant

- “px holds daily BET closes, 2000–2026. Give me the one-day historical VaR 1% and ES 2.5% as positive losses in %, and the 10-day VaR 1%.”

### The assistant's answer

```
import numpy as np
r = 100 * np.log(px).diff().dropna()          # daily log returns, %
loss = -r                                     # loss in % of the position
var1 = np.percentile(r, 99)                  # VaR 1%
es25 = loss[loss >= np.percentile(loss, 99)].mean() # ES 2.5%
var10 = 10 * var1                            # 10-day VaR 1%
```

- “VaR 1% = 3.95%, ES 2.5% = 6.28%, 10-day VaR 1% = 39.46% of the position.”

## C2: Critique an AI Answer [Proposed]: Task

Solution discussed in class

### ▣ Data and assumptions

- ▶ The same BET data as in B1 (2000–2026, daily log returns in %); the code on the previous slide runs without errors, but the answer contains **three** mistakes typical for VaR and ES; everything else is correct.

### ▣ Tasks

- ▶ (a) For each mistake, name the line and explain in one sentence why it is wrong, using the definitions  $\text{VaR}_\alpha = -q_\alpha(r)$  and  $\text{ES}_\alpha = \text{mean loss beyond } \text{VaR}_\alpha$ .
- ▶ (b) Say how you would detect it: a count of observations, a number from B1, or a slide of the lecture.
- ▶ (c) Write the corrected line(s) and recompute the three numbers.
- ▶ Hint: count how many daily losses es25 averages, and how many it should average.

### ▣ Report

- ▶ Log each mistake as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, and the correct version. Model: B1.

### C3: Dynamic (VaR, ES) Models, a Project Seed [Open]: Task (1/2)

#### ▣ Context and data

- ▶ Lecture 7, case study: [Patton, Ziegel & Chen \(2019\)](#) model  $v_t = -\text{VaR}_{\alpha,t}$  and  $e_t = -\text{ES}_{\alpha,t}$  jointly and estimate the parameters by minimising the average FZ0 loss  $L_{\text{FZ0}}(Y, v, e; \alpha) = -\frac{1}{\alpha e} \mathbf{1}\{Y \leq v\}(v - Y) + \frac{v}{e} + \ln(-e) - 1$ ; on the S&P 500, 2000–2016, their one-factor GAS model had the lowest average loss at  $\alpha = 5\%$ , while across the four indices GARCH estimated by FZ0 had the best average rank at  $\alpha = 1\%$  and 2.5% (Table 11).
- ▶ Daily returns in %,  $Y_t = 100(\ln P_t - \ln P_{t-1})$ , from data/market: S&P 500 and Nikkei 225 (from January 1990) and BET (from 1997), each on its own calendar.

#### ▣ Tasks

- ▶ (a) For the S&P 500 and the Nikkei 225 over 1 January 1990 – 31 December 2016, compute the sample VaR and ES at  $\alpha = 1\%$ , 2.5%, 5% and 10% and compare them with Table 6 of the paper.
- ▶ (b) On 1990–1999 at  $\alpha = 5\%$ , estimate three models: the one-factor GAS (Eq. 20,  $\omega = 0$ ) and GARCH(1,1) (Eqs. 25–26,  $\omega = 1$ ), both by minimising the average FZ0 loss with the warm-start procedure of Appendix C, and GARCH(1,1) with FHS residuals estimated by QML; keep the parameters fixed and forecast  $(v_t, e_t)$  one day ahead for 2000–2016.

## C3: Dynamic (VaR, ES) Models, a Project Seed [Open]: Task (2/2)

### ▣ Tasks (continued)

- ▶ (c) Compare the three models by the average out-of-sample FZ0 loss and by DM tests [Diebold & Mariano \(1995\)](#) with HAC errors, and set your numbers next to Tables 9 and 10.
- ▶ (d) Extension: repeat (b)–(c) at  $\alpha = 2.5\%$  with estimation on 2007–2016 and evaluation on 2017–2026, for the S&P 500, the Nikkei 225 and the BET.

### ▣ Report

- ▶ A table per index, level and period: average FZ0 of each model, the DM  $t$ -statistics for every pair, the VaR violation rate and the number of violations.
- ▶ Interpretation (one page): does the ranking of Table 11 at  $\alpha = 2.5\%$  (GARCH estimated by FZ0 ahead of the one-factor GAS) hold in 2017–2026 and on the BET, and how does the number of VaR violations explain what you find?

## Common Pitfalls

- ▣ Mixing sign conventions: a VaR reported as a negative return in one table and a positive loss in another
- ▣ Using the Normal distribution or Cornish–Fisher with daily kurtosis of 10
- ▣ Reporting ES 1% from 500 days without saying it averages 5 numbers
- ▣ Scaling by  $\sqrt{10}$  a series with positive autocorrelation (BET, the BVB basket)
- ▣ Reading money weights as risk weights: a 10% crypto position can carry over a third of the risk
- ▣ Log returns summed across assets: portfolio losses need simple returns

## Take-Home Assignment (Team Project)

- Choose a portfolio of at least three assets not covered in class (BVB stocks, a currency, a crypto-asset)
- Estimate one-day  $\text{VaR}_{1\%}$  and  $\text{ES}_{2.5\%}$  with historical, filtered and extreme-value methods, each with a bootstrap interval
- Split the ES into Euler contributions; from equal weights (long-only, fully invested, historical ES 2.5%, sample-mean expected return), minimise ES subject to the same expected return and aim for a 20% reduction; if it is infeasible, report the minimum attainable ES
- Log every AI error you catch in `AI_ERRORS.md` (at least three per project)
- Deliver a Quantlet folder (code, MetaInfo, charts) and three slides: numbers, interpretation, limitations

## Further Study on Quantinar

### Advanced courses on quantinar.com

- ▣  Measuring Statistical Risk
- ▣  XFG Advanced Methods in Quantitative Finance

### Code and data

- ▣ Seminar notebook (Colab); Quantlets: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_07](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_07)

## References (1/3)

- ▣ Acerbi, C., Tasche, D. (2002). [On the coherence of expected shortfall](#). *Journal of Banking & Finance*, 26(7), 1487–1503.
- ▣ Artzner, P., Delbaen, F., Eber, J.-M., Heath, D. (1999). [Coherent measures of risk](#). *Mathematical Finance*, 9(3), 203–228.
- ▣ Barone-Adesi, G., Giannopoulos, K., Vosper, L. (1999). [VaR without correlations for portfolios of derivative securities](#). *Journal of Futures Markets*, 19(5), 583–602.
- ▣ Basel Committee on Banking Supervision (2019). [Minimum capital requirements for market risk](#). Bank for International Settlements.
- ▣ Chen, S.X. (2008). [Nonparametric estimation of expected shortfall](#). *Journal of Financial Econometrics*, 6(1), 87–107.
- ▣ Christoffersen, P., Gonçalves, S. (2005). [Estimation risk in financial risk management](#). *Journal of Risk*, 7(3), 1–28.
- ▣ Cornish, E.A., Fisher, R.A. (1938). [Moments and cumulants in the specification of distributions](#). *Revue de l'Institut International de Statistique*, 5(4), 307–320.
- ▣ Diebold, F.X., Mariano, R.S. (1995). [Comparing predictive accuracy](#). *Journal of Business & Economic Statistics*, 13(3), 253–263.
- ▣ Efron, B. (1979). [Bootstrap methods: another look at the jackknife](#). *Annals of Statistics*, 7(1), 1–26.

## References (2/3)

- ▣ Engle, R.F., Manganelli, S. (2004). *CAViaR: conditional autoregressive value at risk by regression quantiles*. *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Ferro, C.A.T., Segers, J. (2003). *Inference for clusters of extreme values*. *Journal of the Royal Statistical Society: Series B*, 65(2), 545–556.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). *Statistics of Financial Markets: An Introduction*, 5th ed. Springer.
- ▣ Künsch, H.R. (1989). *The jackknife and the bootstrap for general stationary observations*. *Annals of Statistics*, 17(3), 1217–1241.
- ▣ McNeil, A.J., Frey, R. (2000). *Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach*. *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ McNeil, A.J., Frey, R., Embrechts, P. (2015). *Quantitative Risk Management: Concepts, Techniques and Tools*, revised ed. Princeton University Press.
- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). *Dynamic semiparametric models for expected shortfall (and Value-at-Risk)*. *Journal of Econometrics*, 211(2), 388–413.
- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). *Finite-sample precision limits for expected shortfall forecast comparisons*. *Mathematics*, 14(13), 2316.
- ▣ Rockafellar, R.T., Uryasev, S. (2000). *Optimization of conditional value-at-risk*. *Journal of Risk*, 2(3), 21–41.

## References (3/3)

- ▣ Tasche, D. (2008). [Capital allocation to business units and sub-portfolios: the Euler principle](#). [arXiv:0708.2542](#).